

Some Notes on Languages Accepted by a Turing Machine

Some Definitions

For every Turing machine (TM), T , which runs on strings from the alphabet Σ , we can divide the set of all finite strings over Σ into three disjoint set:

$$\Sigma^* = \text{Accept}(T) + \text{Reject}(T) + \text{Loop}(T).$$

This leads to two possible definitions for the languages that are recognized by Turing machines.

Definition: A language L over alphabet Σ is called **recursively enumerable (Turing-recognizable)** if there is a TM, T , that accepts every string (word) in L and either rejects or loops for every string in L' , the complement of L (i.e. every string in $\Sigma^* - L$).

$$\begin{aligned}\text{Accept}(T) &= L, \\ \text{Reject}(T) + \text{Loop}(T) &= L'\end{aligned}$$

Definition: A language over the alphabet Σ is called **recursive (Turing-decidable)** if there is a TM, T , that accepts every string in L and rejects every string in L' .

$$\begin{aligned}\text{Accept}(T) &= L \\ \text{Reject}(T) &= L' \\ \text{Loop}(T) &= \emptyset\end{aligned}$$

Every recursive language is also recursively enumerable. But there are some languages that are recursively enumerable (r.e.) but not recursive.

Closure Properties of Recursive and R.E. Languages

Theorem 1: The complement of a recursive language is recursive. See the given handout for a proof.

Theorem 2: The union of two recursive languages is recursive. The union of two recursively enumerable languages is recursively enumerable.

Theorem 3: If a language L and its complement L' are both recursively enumerable, then L (and hence L') is recursive.

Theorems 1, 2, and 3 have an important consequence. Let L and L' be a pair of complementary languages. Then either:

- (1) both L and L' are recursive,
- (2) neither L nor L' is r.e., or
- (3) one of L and L' is r.e. but not recursive; the other is not r.e.

Proof:

Using the following abbreviations we construct the table given below.

R = Recursive

RE = Recursively enumerable

NRE = Not recursively enumerable

Case	1	2	3	2	4	5	3	5	6
L	R	RE	NRE	R	RE	NRE	R	RE	NRE
L'	R	R	R	RE	RE	RE	NRE	NRE	NRE

Case 1 is (1), case 6 is (2) and case 5 is (3). Cases 2 & 3 are impossible (Theorem 1). Case 4 is impossible (Theorem 2).

In class I discussed how to encode a TM using alphabet $\{a, b\}$. Let's call this language the Code Word Language (CWL).

CWL = the language defined by $(a^+ b a^+ b (a + b)^5)^*$

Since the code for every TM is a string of **a**'s and **b**'s, we might ask what happens if this string is run as input on the very TM it stands for? We shall feed each TM its own code word as input data. Sometimes it will crash, sometimes loop, sometimes accept.

Let us define the language Alan as follows:

Alan = {all the strings in CWL that are *not* accepted by the TM's they represent or that do not represent any TM}

We now prove that the language Alan is not recursively enumerable. This is a proof by contradiction. Assume that Alan is r.e.. In that case there would be some TM that would accept all the strings in Alan. Let's call this Turing machine T. Let's denote the code word for T as Code(T). Now, ask the following question:

Is Code(T) a string in the language Alan or not?

There are only two possibilities: Yes (case 1) or No (case 2).

Case 1: Assume $\text{Code}(T)$ is in Alan. Since Alan contains no encoding of TM's that are accepted by their respective TM's, $\therefore \text{Code}(T)$ cannot be in Alan. Hence, a contradiction.

Case 2: Assume $\text{Code}(T)$ is not in Alan. Since Alan contains all the encodings of TM's that are not accepted by their respective TM's, $\therefore \text{Code}(T)$ must be in Alan. Again, a contradiction.

Let us define another language called M.

$$M = \{\text{all the strings in CWL that are accepted by their corresponding TM}\}$$

It can be proved that M is recursively enumerable.

Some Conclusions about M Being Recursively Enumerable

1. CWL is regular because it is defined by a regular expression.
2. The complement of CWL in $\{a, b\}^*$, called CWL' , is also regular.
3. CWL' is r.e..
4. $\text{CWL}' + M$ is r.e..
5. The complement of $(\text{CWL}' + M)$ is Alan, since Alan is all code words not in M.
6. But Alan is not r.e.. Therefore, we have proved the following:

The complement of a recursively enumerable language may not be recursively enumerable.

We know that M is a r.e. language, but is it recursive? CWL' is regular and therefore recursive. The union of two recursive languages is also recursive. Therefore, if M is recursive so is:

$$M + \text{CWL}' = \text{Alan}' = \text{the complement of Alan}$$

But, this would mean that Alan is recursive, which is not true, because Alan is not even r.e.. Therefore, we have proved:

There are recursively enumerable languages that are not recursive.